

Project Proposal

Group 30

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1 Overview

This project focuses on training a machine learning model to accurately segment brain tumors from MRI scans using a 2D U-Net convolutional neural network. Brain tumor classification involves identifying tumor regions within an MRI image.

Automated systems have the potential to support doctors by providing an initial segmentation that can be reviewed by the doctor, helping reduce the time for manual segmentation. Further, automated systems can in many cases act as a proof-reader, providing an additional safeguard towards any false-negatives.

Brain tumor classification is a challenging machine learning problem since tumors can vary significantly in appearance between subjects. Further, tumor tissue may also be similar visually to surrounding healthy tissue, or may only take up a small region of the MRI image.

2 Solution

U-Net is a convolutional neural network (CNN) architecture designed for image segmentation tasks. This architecture consists of an encoder and decoder. The encoder repeatedly applies convolutions and downsamples the image. It learns increasingly abstract features. The decoder upsamples these features back up to the original resolution of the input. Its job is to figure out the exact location of features. In the context of our project, encoded early layers will detect edges and features, while deeper layers can learn patterns associated with tumors. [1]

Each MRI scan consists of four images: T1, contrast-enhanced T1 (T1CE), T2, and T2-FLAIR. These images will be stacked as separate input channels, allowing the model to use information from each image concurrently in its prediction.

The target labels will consist of the already labeled segmentation masks provided with the dataset. Therefore, as the labels are pixel level, the model will also perform pixel-wise classification, assigning each pixel in the U-Net slice to either the background or one of the annotated tumor regions.

The model will be built based off a U-Net architecture, which is a tried and proven workflow designed specifically for biomedical image classification tasks. PyTorch will be used for implementation and training/validation.

References

- [1] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. *CoRR*, abs/1505.04597, 2015.